

Random Forests: Variance, Correlation, and the Correlation Floor

1. Objective

Understand why random feature selection in Random Forests reduces variance and improves ensemble performance.

The central quantity of interest is:

$$\text{Var}(\bar{T})$$

where:

$$\bar{T}(x) = \frac{1}{M} \sum_{i=1}^M T_i(x)$$

is the ensemble prediction at a fixed input x .

2. Exact Variance Decomposition

Start with:

$$\text{Var}(\bar{T}) = \text{Var}\left(\frac{1}{M} \sum_{i=1}^M T_i\right) = \frac{1}{M^2} \text{Var}\left(\sum_{i=1}^M T_i\right)$$

Expand the variance of a sum:

$$\text{Var}\left(\sum_{i=1}^M T_i\right) = \sum_{i=1}^M \text{Var}(T_i) + \sum_{i \neq j} \text{Cov}(T_i, T_j)$$

Thus:

$$\text{Var}(\bar{T}) = \frac{1}{M^2} \left(\sum_{i=1}^M \text{Var}(T_i) + \sum_{i \neq j} \text{Cov}(T_i, T_j) \right)$$

3. With Exchangeability (Simplified Model)

If trees are exchangeable:

$$\text{Var}(T_i) = \sigma^2$$

$$\text{Cov}(T_i, T_j) = \rho\sigma^2$$

Then:

$$\text{Var}(\bar{T}) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2$$

As $M \rightarrow \infty$:

$$\text{Var}(\bar{T}) \rightarrow \rho\sigma^2$$

The correlation floor is now $\rho\sigma^2$.

4. What Drives the Correlation Floor?

The floor equals the **average pairwise covariance** between trees.

Correlation arises from:

1. Shared underlying signal ($f(x)$)
2. Dominant predictors
3. Redundant (correlated) features
4. High signal-to-noise ratio
5. Structural similarity of the tree-growing algorithm

Even with randomness, trees approximate the same function, so covariance cannot be zero.

5. Why Random Feature Selection Helps

At each split, sample m out of p features.

Probability a dominant feature is available:

$$P(\text{strong feature available}) = \frac{m}{p}$$

If m is small:

- Strong predictors are often excluded
- Trees follow alternative split paths
- Tree structures diversify
- Correlation ρ between trees decreases

Thus:

$$\text{Var}(\bar{T}) \approx \rho\sigma^2$$

Reducing m reduces the correlation floor.

6. Bias--Variance Tradeoff

However, smaller m :

- Weakens individual trees
- May increase per-tree variance
- May increase bias

Approximate ensemble MSE:

$$MSE(m) = Bias(m)^2 + \rho(m)\sigma^2(m) + \frac{1 - \rho(m)}{M}\sigma^2(m)$$

For large M :

$$MSE(m) \approx Bias(m)^2 + \rho(m)\sigma^2(m)$$

Choose m to balance strength and correlation.

7. Why $m \approx \sqrt{p}$?

Heuristic compromise:

- Ensures m/p decreases with dimension
- Reduces probability dominant predictors dominate all trees
- Still allows enough features per split to maintain strength

It balances reduction in covariance against increase in bias.

8. Final Logical Flow

1. Under exchangeability, ensemble variance depends on σ^2 and ρ .
 2. As forest size grows, variance converges to $\rho\sigma^2$.
 3. Correlation arises from shared signal and dominant predictors.
 4. Random feature selection reduces shared structure across trees.
 5. Reduced covariance lowers the correlation floor.
 6. Optimal m balances tree strength and decorrelation.
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Core Insight:

Random forests work because they reduce the average covariance between trees faster than they weaken individual trees.